

Unlocking the Diversity Buffer: Enhancing Generalization in Abstraction-Learned Policies

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1 Objective

The primary objective of this project is to investigate and mitigate "mode collapse" in Large Language Models (LLMs) optimized via Reinforcement Learning (RL). While standard RL training typically causes models to converge on a narrow set of high-reward responses, the **RLAD (Reinforcement Learning from Abstractions and Discovery)** framework (Qu et al., 2025) demonstrates that training with procedural hints allows models to internalize complex strategies. This project seeks to extend these findings by shifting the focus from accuracy to distributional breadth. We aim to determine if the internalized strategies in RLAD-trained models provide a "Diversity Buffer"—a latent ability to generate semantically varied solutions—and to explicitly optimize this buffer using a **Semantic Diversity Reward** in a two-player RL setting.

This research is relevant because real-world problem-solving requires open-ended reasoning; ensuring diversity in generated solutions prevents brittle decision-making and significantly improves the likelihood of discovering correct trajectories on complex, unseen problems.

2 Related Work

Recent advancements in test-time scaling have highlighted the importance of in-context exploration and extrapolation. Specifically, the **e3 framework** Setlur et al. (2025) demonstrates that training models to test multiple hypotheses and chain asymmetric skills (such as verification and generation) enables models to extrapolate their performance when given longer test-time compute budgets. Complementing this, the **Reasoning Cache (RC) framework** Wu et al. (2026) exploits the asymmetry between response generation and summarization to construct reasoning chains that continually improve over long horizons. However, while these approaches successfully scale reasoning performance, the underlying exploration still risks converging onto singular, repetitive reasoning paths without explicit diversity constraints.

Our project builds directly on the RLAD framework, which treats reasoning as a co-evolutionary two-player game: an **Abstraction Generator** proposes high-level procedural summaries, and a **Solving Network** executes them. A key finding of the RLAD paper is a distinct **generalization phenomenon**: although the solving network is trained to utilize and rely on abstractions during the two-player phase, it significantly outperforms standard RL baselines when inferring independently—using the solving network alone without any abstraction input.

The authors measure this performance increase, but note that understanding and amplifying it remains an open question for future work. Our project directly addresses this gap. We plan to measure and characterize the underlying diversity of this generalization effect in Phase 1, and subsequently introduce a novel Semantic Diversity Reward framework in Phase 2 to actively amplify this discovery capability and prevent mode collapse.

3 Technical Outline

Our approach treats the challenge of mode collapse as an explicit reward-shaping problem within the RLAD two-player game. Our novel technical contribution is the integration of a semantic novelty penalty into the abstraction generation loop to ensure the model explores varied lines of reasoning.

3.1 Phase 1: Quantifying the "Diversity Buffer" in Independent Inference

We will investigate whether the generalization effect observed in the RLAD solving network extends to reasoning breadth. Because highly challenging competition datasets can yield low success rates—making it difficult to analyze diversity among correct paths—we will conduct our primary evaluation using the MATH and GSM8K benchmarks, where the higher frequency of correct answers

ensures a statistically meaningful sample of successful solution traces. We will sample k distinct solutions from both a standard RL baseline and the RLAD solving network (operating without abstractions) to quantify structural redundancy (using Self-BLEU) and semantic variance (using SBERT embeddings).

3.2 Phase 2: Implementing the Semantic Diversity Reward

To prevent the Abstraction Generator from collapsing into a few "safe" hints, we will introduce a diversity-aware objective. The Abstraction Generator (π_{θ}^{abs}) will receive a composite reward:

$$R_{total} = R_{accuracy} + \lambda \cdot \text{Novelty}(A_t \mid \mathcal{B})$$

where A_t is the current abstraction and $\mathcal{B}$ is a memory buffer of abstractions previously successful for that specific problem. The novelty score will be derived from the semantic distance between the current abstraction and the cluster centroids of previous successful abstractions stored in the buffer.

3.3 Phase 3: Discovery Analysis & Evaluation

We will evaluate whether the Diversity-Augmented RLAD discovers a higher number of unique, valid "algorithmic procedures" (e.g., different mathematical approaches to the same MATH or GSM8K problem) compared to the original RLAD. An ablation study will compare the solving network's independent performance vs. its abstraction-guided performance to see if the Diversity Reward further enhances generalization. Finally, we will analyze the Pareto frontier between reasoning accuracy and solution diversity to determine if incentivizing breadth harms or helps overall success in multi-step reasoning tasks.

4 Team Contributions

- **Chung:** Leads Phase 1. Chung will focus on designing and executing the evaluation of independent inference on the RLAD solving network, gathering responses from the baseline and RLAD models, and calculating structural and semantic diversity metrics such as Self-BLEU and SBERT cosine similarities.
- **Anna:** Leads Phase 2. Anna will design and implement the semantic diversity reward mechanism, establishing the memory buffer of previous abstractions and integrating the novelty penalty into the Abstraction Generator's training objective.
- **Chung and Anna:** Jointly execute Phase 3. Both team members will collaborate on the discovery analysis, conduct ablation studies comparing the abstraction-guided versus independent performance, and analyze the Pareto frontier between diversity and reasoning accuracy.

References

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